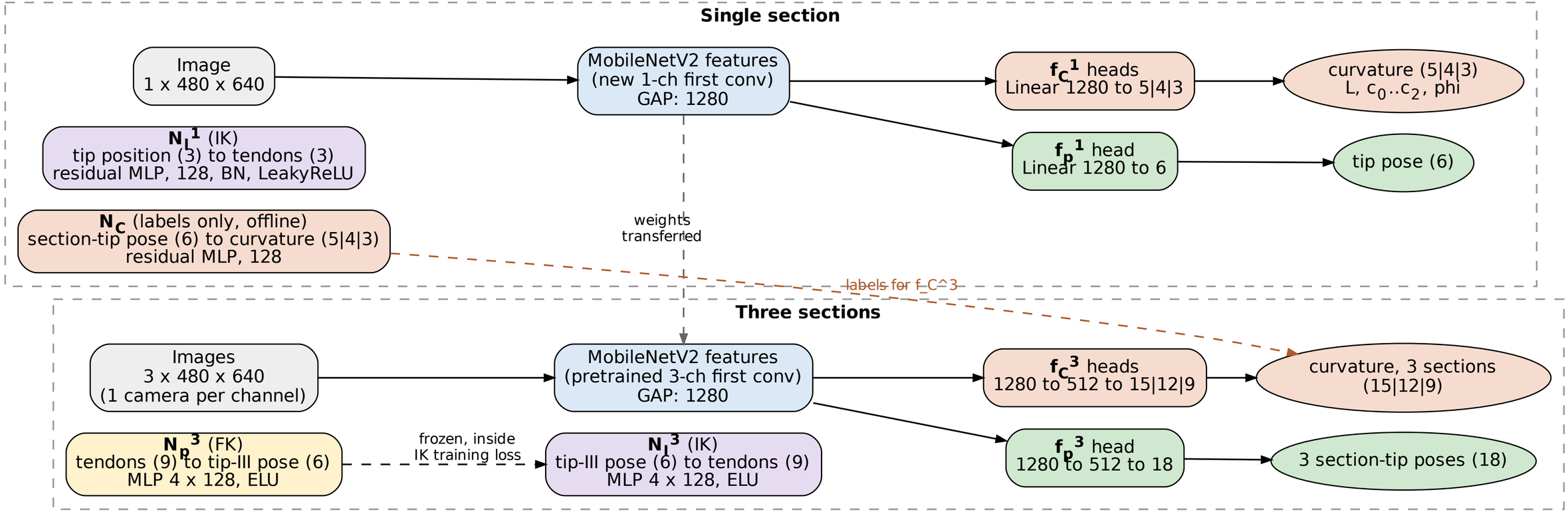
Single section
Image
1 x 480 x 640
MobileNetV2 features
(new 1-ch first conv)
GAP: 1280
$\mathbf{f_C}^1$ heads
Linear 1280 to 5|4|3
curvature (5|4|3)
L, c_0..c_2, phi
$\mathbf{N_l}^1$ (IK)
tip position (3) to tendons (3)
residual MLP, 128, BN, LeakyReLU
$\mathbf{f_p}^1$ head
Linear 1280 to 6
tip pose (6)
$\mathbf{N_C}$ (labels only, offline)
section-tip pose (6) to curvature (5|4|3)
residual MLP, 128
weights
transferred
labels for f_C^3
Three sections
Images
3 x 480 x 640
(1 camera per channel)
MobileNetV2 features
(pretrained 3-ch first conv)
GAP: 1280
$\mathbf{f_C}^3$ heads
1280 to 512 to 15|12|9
curvature, 3 sections
(15|12|9)
$\mathbf{N_p}^3$ (FK)
tendons (9) to tip-III pose (6)
MLP 4 x 128, ELU
frozen, inside
IK training loss
$\mathbf{N_l}^3$ (IK)
tip-III pose (6) to tendons (9)
MLP 4 x 128, ELU
$\mathbf{f_p}^3$ head
1280 to 512 to 18
3 section-tip poses (18)